STA380 Group Project Proposal

Permutation Test on World Population Development Groups (2023)

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The following proposal is an example of using an "original" topic that has not already been suggested in the main project document. I plan to teach the material in this example in the last few weeks of the term. Of course, since I am providing this example (with all raw codes), you are not permitted to use MCMC methods for your project unless you want to do something more advanced and outside the scope of the course. Coloured text represents additional comments.

I used an Undergraduate Project Capstone Proposal Template to write this example

1 Project Topic

Permutation Test between Developmental Groups from the World Population Prospects (2024)

Our group proposes to perform a permutation test to determine if there is a statistically significant difference in chosen demographic indicators (such as Life Expectancy, Birth Rate, or Mortality Rate) between different developmental groups (e.g., Developing vs. Developed regions[1]).

2 Simulation vs. Dataset

This project will use a **dataset** sourced from the United Nations [United Nations, Department of Economic and Social Affairs, Population Division, 2024].

- **Source:** United Nations, Department of Economic and Social Affairs, Population Division.
- **Dataset Name:** World Population Prospects 2024.
- **Link:** https://population.un.org/wpp/

We will use the "Estimates" worksheet from the WPP 2024 dataset, which contains reconstructed historical data from past censuses and surveys, ensuring our permutation test is performed on facts rather than projections of the future (such as the "Medium Variant", "Low Variant", and "High Variant" sheets).

To capture the current state of global development, we will filter the dataset for the year 2023, which represents the most recent year of estimated historical data.

The processed `Clean_data.csv` file containing the relevant indicators and developmental groups has been provided within the Proposal sub folder.

[1] Countries were classified into 'Developed' and 'Developing' groups according to the United Nations Statistics Division M49 Standard. 'Developed' includes Europe, Northern America, Australia, New Zealand, and Japan; all other regions are classified as 'Developing'.

3 Project Details

We will be using 3 introductory MCMC methods to simulate the bivariate Gaussian distribution:

- An independence Metropolis–Hastings algorithm
- A random walk Metropolis–Hastings algorithm
- A Gibbs Sampler

The following outputs will be provided:

- Scatter plot of the samples generated
- Histogram and Q-Q plots to assess marginal normality
- The generated mean and the covariance matrix

Aside: if extra details mostly pertain to the next section, then you can just write them below instead of repeating yourself.

4 User Inputs (Shiny Components)

Users will be able to modify the following:

1. Setting the seed
2. Change the parameters, specifically, for $i \in \{1, 2\}, \mu_i, \sigma_i$ and correlation ρ.
3. Select the information presented:
   - the mean and covariance matrix,
   - histogram and Q-Q plot for marginal normality,
   - and the scatterplot of the samples generated.
4. Be able to choose what information is presented, i.e., they can look at results from just 1 MCMC method, 2 MCMC methods, or all three at once.
5. For the plots, be able to modify the colours.

5 Generative AI Usage Declaration

Generative AI was used in this proposal to convert the provided LaTex proposal example to a R Markdown format.

References

W. Chang, J. Cheng, J. Allaire, C. Sievert, B. Schloerke, G. Aden-Buie, Y. Xie, J. Allen, J. McPherson, A. Dipert, and B. Borges. shiny: Web application framework for r, 2025. URL https://CRAN.R-project.org/package=shiny. R package version 1.12.1.

R Core Team. R: A language and environment for statistical computing, 2024. URL https://www.R-project.org/. Vienna, Austria.

United Nations, Department of Economic and Social Affairs, Population Division. *World Population Prospects 2024, Online Edition*. United Nations, 2024. URL https://population.un.org/wpp/.

United Nations Statistics Division. Standard country or area codes for statistical use (m49), 2024. URL https://unstats.un.org/unsd/methodology/m49/. See section 1 (Project Topic) for Note on developed and developing regions.